

SYNOPSIS

On 24 June 2022, while alongside at King George Dock, Hull, England, the crew of the UK registered fishing vessel *Kirkella* lost control of its propulsion system and the vessel collided with the unmanned tug *Shovette*, which was moored ahead of *Kirkella*. During the collision *Shovette*'s hull and starboard fuel tank were breached by *Kirkella*'s bulbous bow. The tug partially sank, which resulted in pollution of about 7,000 litres of marine diesel oil being released into the dock. *Kirkella* was not significantly damaged and there were no injuries.

The investigation found that the pitch levers for *Kirkella*'s propulsion control system were mismatched between the bridge and engine control room when control was transferred. The classification society's interpretation of the requirement that the control system should prevent the propulsion from altering significantly when transferring control between stations allowed the pitch levers to be mismatched when changing control from the bridge to the engine control room.

Kirkella's operating company, UK Fisheries Ltd, has introduced measures to help stop a reoccurrence until the system can be retrofitted with prevention interlocks. The vessel's classification society, Det Norske Veritas, has been recommended to propose to the International Association of Classification Societies that it reviews its Unified Requirement for remote control of propulsion systems. The system support company, Kongsberg Maritime, has been recommended to issue a service letter and advise its customers of available options to prevent propulsion from altering significantly when transferring control.

1.2 NARRATIVE

1.2.1 Events leading to the collision

On 23 June 2022, *Kirkella* was on passage between its fishing grounds and Hull, England. During the passage the ship's engineers carried out engine maintenance that required propulsion control to be transferred from the bridge to the engine control room and the vessel to be at full speed. During the trial the pitch control lever in the engine control room (ECR) was set to full ahead. When control was passed back to the bridge on completion of the engine maintenance the ECR pitch control lever was left in the full ahead position.

On 24 June, *Kirkella* returned from its fishing grounds off northern Norway and was piloted up the Humber river to the lock at King George Dock, Hull¹. The master, first officer (1/O), and a pilot were on the bridge. The wind was south-south-westerly force 2 with good visibility.

At 0531, *Kirkella*'s master manoeuvred the vessel from the lock towards the berth using the starboard helm station. The master then berthed the vessel starboard side alongside 12 Quay and adjacent to shed 11 (**Figure 1**).

Image courtesy of [Associated British Ports](#)

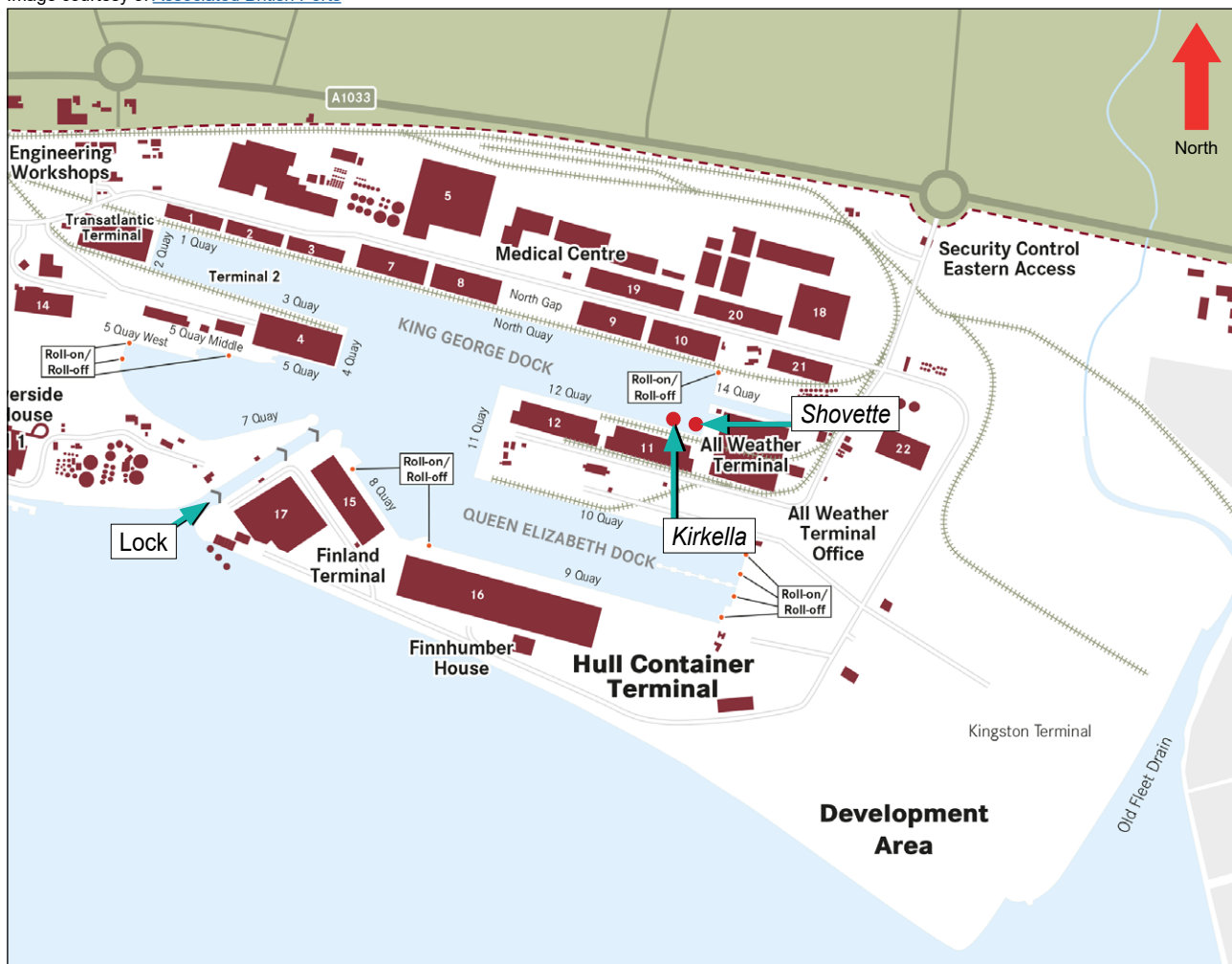


Figure 1: King George Dock

¹ Operated by Associated British Ports (ABP).

At 0557, *Kirkella*'s crew sent mooring lines ashore to the port's line handlers, who secured them to the bollards. The master positioned the vessel to make sure its cargo of frozen fish could be discharged through the vessel's cargo door onto the quayside. By 0606, the vessel was secure with two head lines and two spring lines forward and two stern lines and two spring lines aft. The line handlers departed shortly afterwards.

At 0611, *Kirkella*'s master passed control of the vessel's propulsion system from the starboard wing control station to the centre console position (**Figure 2**). At the same time, the crew deployed the vessel's gangway to the quayside. Once the master was content that the vessel was secured in position control of the propulsion system was passed to the ECR (**Figure 3**) so that the main engine could be shut down.

The first engineer (1/E) was sitting in the ECR working at a computer. The 1/E heard the propulsion command change alarm and reached across from their chair and pressed the 'ECR' then 'Accept' buttons on the touchscreen to accept control (**Figure 4**). Eighteen seconds later, *Kirkella* started to move ahead.

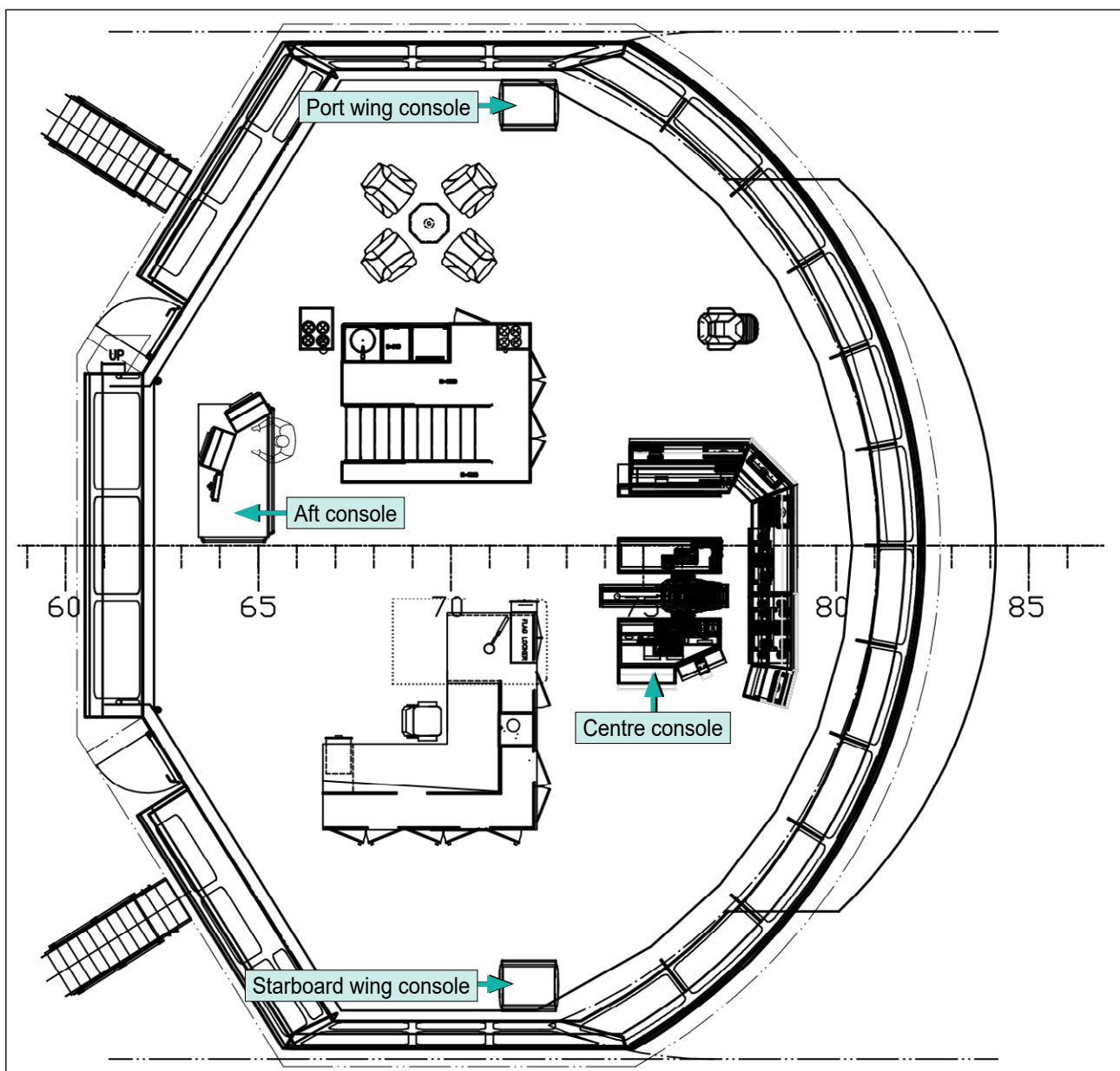


Figure 2: Bridge manoeuvring control consoles



Figure 3: The engine control room



Figure 4: Engineer reaching across to the propulsion control touchscreen

On the bridge, the 1/O was shutting down the bridge equipment when they noticed the vessel moving ahead and raised the alarm. The master reacted to the 1/O's exclamation and saw that the vessel was moving ahead. Shortly afterwards, the two stern lines parted and the forward spring lines surged on the mooring bitts.

The master went to the centre console and telephoned the ECR, instructing the 1/E to stop the engine. At about the same time, the master pushed the emergency stop button on the centre console, which declutched the engine. Moments before the master's telephone call, having sensed a change in the main engine load, the 1/E had checked the engine control console and observed that the pitch was 85%.

As *Kirkella* continued to move forward, two further mooring lines parted and the gangway was dragged until it fell from the quayside. Ten seconds after the master had declutched the engine, *Kirkella* collided with the moored and unmanned harbour tug *Shovette* (**Figures 5a to 5c**), which was berthed adjacent to the all-weather terminal at the end of the King George Dock (**Figure 1**). *Kirkella* bounced off the tug and came to rest with three forward mooring lines and one aft mooring line still intact.

SECTION 2 – ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 OVERVIEW

Kirkella collided with *Shovette* due to the propeller pitch increasing after propulsion control was passed from the bridge to the engine control room. The vessel had no established procedures for the transfer of propulsion control and the crew had made assumptions based on local practices. The analysis will examine the factors influencing the loss of control of the propulsion system and other circumstances leading to the collision.

2.3 LOSS OF CONTROL

2.3.1 Propulsion control change

After *Kirkella*'s crew had conducted the engine maintenance at sea, the ECR pitch control lever being left at full ahead did not present any problem, until control was passed back to the ECR once the vessel was secured alongside.

When the master requested that the ECR took control of the propulsion using the system panel there was no telephone communication with the 1/E, which could have focused their actions. Certainly, a telephone call would have required the 1/E to leave their chair at the computer and look directly at the propulsion control screen. There was no procedure in place to ensure that confirmation of pitch lever positions was communicated before control was handed over.

Changing control station was a routine matter and the bridge team had developed a process of zeroing the bridge stations' levers before control changeover, but this was done with people in sight and hearing of one another, or by one person changing control, thus minimising the chance of lever mismatch. However, the ECR was remote from the bridge and, although the system display indicated the propeller pitch setting, it did not provide an audible or overt visual alert to highlight when the bridge and ECR levers were mismatched. Consequently, without communication between the two control stations, neither was fully aware of what the other was doing and there was no procedure to ensure that lever matching occurred.

2.3.2 Acceptance of control in the engine control room

The 1/E was at the computer workstation completing logs at the end of their shift. As it was in the idle period between the vessel's arrival alongside and the bridge declaring 'finished with engines', there was no need for focused machinery monitoring.

The sounding of the control change alarm had become routine because the 1/E had completed the acceptance process many times before. With no telephone communication from the bridge to advise of changeover, the 1/E had no reason to stand in front of the propulsion control screen and pitch control lever (see **Figure 4**) and did not expect anything to be different on this occasion. This was evident from the 1/E's immediate return to administrative duties until the telephone call from the

bridge. Standing by the telephone might have caused the 1/E to look at the pitch lever but only a formal check instigated by the bridge, or by a practiced procedure, would have ensured the 1/E was fully engaged in the changeover process.

2.3.3 Response to loss of control

Kirkella's 1/O realised that the vessel was moving ahead and immediately raised the alarm. The increase in *Kirkella's* propeller pitch accelerated the vessel forward until some of its aft mooring lines parted, and one stern line and the forward springs surged on their bitts. This allowed *Kirkella* to further accelerate until it struck *Shovette*. The master responded immediately by telephoning the ECR, but had assessed the ship was gathering speed and so declutched the engine.

It could be viewed that instructing the 1/E to put the engine to full astern might have affected the outcome, but at that stage neither the master nor the 1/E knew what the problem was and reversing pitch would have taken several seconds; only 10 seconds elapsed between declutching and the collision. The master's action to declutch the engine was an appropriate response in the circumstances.

The declutching of the engine and that some of *Kirkella's* mooring lines remained intact, while not enough to prevent the collision, certainly lessened the consequences.

2.4 FATIGUE

The 1/E had been working 12-hour night shifts before the accident and was at the end of their watch. Although the 1/E might not have felt tired, studies show that the sleep debt and disruption to circadian rhythm¹⁰ caused by this type of shift pattern could significantly affect alertness and concentration. The 1/E had been working at the computer before the propulsion control change alarm sounded; the distraction of administrative duties, coupled with previous experiences of control changeover and the likely effects of fatigue at the end of the shift, resulted in the 1/E not fully engaging with the process when the control change alarm sounded.

2.5 SAFETY MANAGEMENT

Kirkella's SMS encompassed the requirements of MGN 596 (F), with the exception of training. The procedures were generic and did not cover vessel-specific equipment requirements. The omission of training requirements was notable, given these were fundamental to safe operations; however, it is unlikely that this was a contributory factor in the accident given there was no onboard procedure for propulsion control changeover.

The focus of an SMS was to identify and mitigate general risks associated with critical processes and activities that could potentially lead to accidents or regulatory transgressions. Onboard procedures could be developed for such operations to make sure they were performed consistently and safely. However, less critical or routine operations that might not have dedicated procedures were still expected to be carried out following general safety guidelines and practices. Propulsion control changeover was a routine activity that had not been identified as having a critical effect on safety due to the knowledge of those on board. It was unlikely that an external audit of the SMS would have exposed any weaknesses in the finer details of operational functions.

¹⁰ The pattern of biological processes that occurs regularly at about 24-hour intervals, also known as the sleep/wake cycle. Further information can be found at <https://www.sleepfoundation.org/circadian-rhythm>

SECTION 3 – CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The engine control room pitch control lever had been left in the 'full ahead' position following engine maintenance conducted while the vessel was at sea. [2.3]
2. *Kirkella's* engineer accepted control without checking that the pitch control lever was set at zero. With no interlocks to prevent changeover with mismatched levers, control was taken at the ECR control station with its pitch control lever at 'full ahead' and the propeller pitch increased to 85% before the master pressed the emergency declutch button. [2.3]
3. There was no onboard procedure for transfer of propulsion control and training requirements for its use had been omitted from *Kirkella's* safety management system. [2.5]
4. As fitted to *Kirkella*, the Rolls-Royce Helicon-X3 propulsion control system did not align to the standard of UR M43.12, which required a means to prevent significant alteration of the propelling thrust when transferring control. [2.6, 2.7]
5. DNV's class requirement addressing the standard in UR M43.12 introduced ambiguity, which in turn allowed the Rolls-Royce Helicon-X3 propulsion control system to be approved for use in *Kirkella*. [2.6, 2.7]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. *Kirkella's* crew did not realise that the propeller pitch had increased until the vessel moved forward. [2.3]
2. The increase in pitch accelerated *Kirkella* forward and parted several mooring lines, which allowed it to continue the vessel's movement along the quay until it collided with *Shovette*. [2.3]
3. It is likely that the effects of fatigue after a long shift coupled with the absence of a robust procedure led to the engineer paying insufficient attention to the change of propulsion control to the ECR. [2.4]

3.3 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The Rolls-Royce Helicon-X3 propulsion control system had been fitted to other vessels with remote control stations in the engine control room. Those systems might also not align with the requirement of UR M43.12. [2.6]

Safety lessons

1. The propulsion control system on *Kirkella* was not fitted with interlocks to prevent a mismatched propeller pitch lever position when control was transferred. Over 4,000 Rolls-Royce Helicon-X3 systems have been supplied to the industry, most of which were not fitted with optional interlocks, which were not required. Retrofitting can be undertaken by Kongsberg Maritime, as customer support provider.
2. Robust shipboard practices are essential to maintain propulsion control during handover, regardless of whether system interlocks are fitted. Documented procedures should contain a requirement for the operator of the sending station and the receiving station to check that propulsion systems pitch settings are aligned at the time of transfer.
3. To reduce the risk of propulsion thrust being applied inadvertently while alongside it is advisable to declutch engines before transfer of control.

This flyer and the MAIB's investigation report are posted on our website: www.gov.uk/maib

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Publication date: June 2024

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NOTE

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